Proposed changes for [A91], agents that lock at a workstation

12 November 2026, version 2

How to read this

Part 1 says what happens and what was decided, in plain words. Part 2 shows the four places in the thesis that would change. Text in green is proposed and not in the thesis yet. Black text is already there. Italic text is a mockup instruction in the Experiments chapter. Part 3 lists the questions for you. The rationale is entry [A91] in `GAPS.md`.

Part 1. What happens and what was decided

The lock

A workstation hangs off an aisle by a single edge. When that aisle is one lane wide, the agents bringing lines to the workstation queue on it. The agent at the workstation wants to step back onto the aisle, and the first agent in the queue wants the workstation. That is a swap, so conflict resolution makes both wait. The agents behind close the only way out, so the wait lasts until an agent chooses to back out. The section around that workstation stops. The rest of the fleet keeps working.

How often

Over the 258 warehouse seeds of each fleet, a section whose four workstations all hang off a one-lane aisle occurs in 35 seeds at 8 agents (14 %) and 70 seeds at 32 agents (27 %). Two of the six test seeds, 200000 and 200005, have it at both fleet sizes.

Token Passing does not lock. It plans whole paths and keeps each vertex free around every arrival, and every warehouse is well-formed. So the lock is a limitation of the learned controllers on this family of layouts.

What was decided (11 November)

- **No new rule.** A cap on open tasks per workstation was discussed and dropped. It would change the assignment rule every controller shares, and it fights the order model, where all lines of an order go to one workstation. van Knippenberg et al. (2021) have no such rule either.
- **No new layout.** Two-lane aisles would remove the lock, but mean a new warehouse sweep with Token Passing.

- **Train as it is.** Whether a controller learns to back out becomes part of what is measured.
- **Report it.** The number of locked test seeds per controller and storage rule, a threat to validity, and under RQ4 whether messages make the lock rarer.

What messages did so far

One small test only, at 8 agents with congestion-aware storage, three test seeds, checkpoint 8. With the rule for a locked seed you chose (Part 3), two of the three seeds lock without messages and none with them. The fleet with messages also finished more tasks (91 against 43, 155 against 107 and 201 against 143). Version 1 of this PDF said messages did not prevent the lock. That used a looser test, one agent standing still for 50 steps, and is corrected here. Three seeds are too few to claim more, so the full runs decide.

Part 2. The text

2.1 Implementation, "When moves still clash"

The new paragraph follows the one that ends with the circle of waiting agents.

> An agent is never ready when the agents it waits on form a circle, each wanting the next one's vertex. A swap is the shortest such circle. These agents wait, since none of them can move first.
>
> Such a circle can last. The clearest case is a workstation on an aisle one lane wide. The agent at the workstation wants the aisle vertex in front of it, and the agent queuing there wants the workstation. The agents queuing behind close the only way out, so the wait ends only when an agent chooses to back out. No rule is added to break it, because a rule would change the assignment that every controller shares, and van Knippenberg et al. have none either. Whether the learned controllers learn to back out is part of what is measured. Token Passing plans ahead and does not meet this circle.

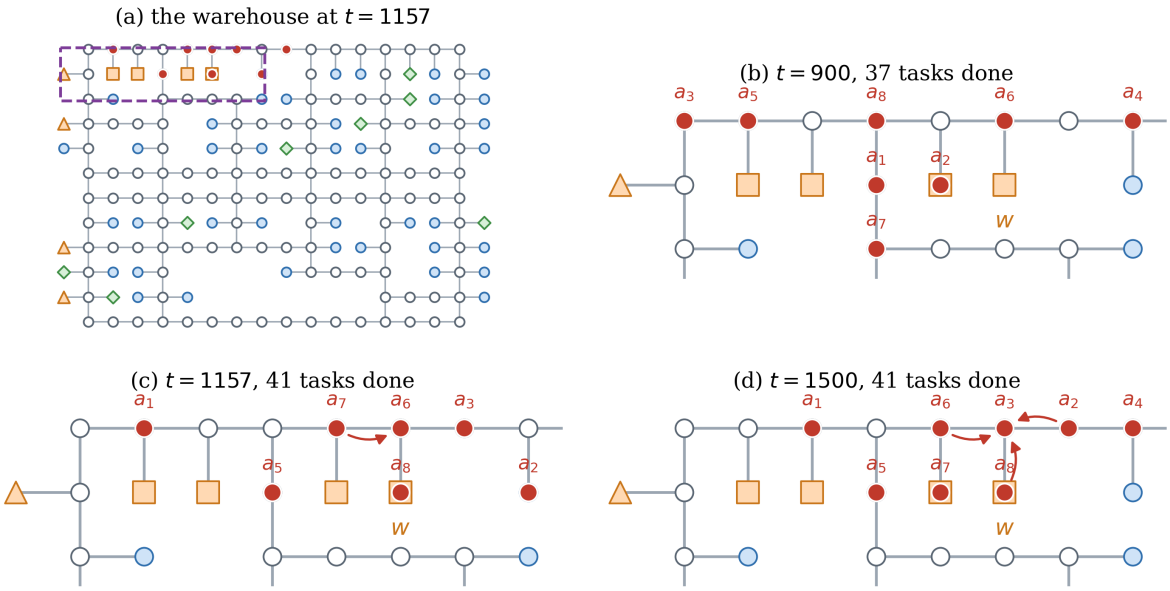


Figure 1: A lock at workstation w in test seed 200000 with eight agents. Arrows show the move an agent asks for.

Figure 1 shows such a lock in a test warehouse with eight agents. All four workstations of one section hang off the outer aisle, which is one lane wide. At $t = 900$ the work still flows, and six agents are on their way to the same workstation w. At $t = 1157$ agent a_8 has delivered at w and must leave it for its next shelf. Agent a_6 stands on the only vertex out and waits to enter w, and a_7 and a_3 queue behind it. Neither a_8 nor a_6 can move first. Seven of the eight agents are caught around this one workstation, so the whole fleet finishes only two more tasks in the last 843 steps of the episode.

2.2 Experiments, Results, "Keeping up"

Keeping up. *Which conditions pass the keep-up check. Report the backlog trace of one condition that falls behind, so the reader sees what failing the check looks like. Give the number of test seeds that lock, per controller and storage rule. A seed locks when at least two agents each hold the same task for 500 steps or more at the same time, a quarter of the episode. A locked seed counts as not keeping up.*

2.3 Experiments, Results, the communication paragraph

RQ5, communication (conditional). *Only if reached. Gain of the decentralised controller with messages over the same controller without. Also the number of locked test seeds with and without messages.*

The paragraph still says RQ5. Since [A83] communication is RQ4. Renaming the Results headings is a separate change and not part of this one.

2.4 Experiments, Discussion, "Threats to validity"

A new item in the list.

- Agents can lock at a workstation on an aisle one lane wide (Section \ref{sec:method:resolution}). A section with all its workstations on such aisles occurs in 14 % of the warehouses at 8 agents and 27 % at 32 agents, and in 5 of the 20 test warehouses at 8 agents. Token Passing does not lock, so in these warehouses the gap between it and the learned controllers is partly the layout.

Part 3. Decided, and what is left

Your answers of 12 November.

1. **A seed is locked** when at least two agents each hold the same task for 500 steps or more at the same time. The figure's seed locks from $t = 1157$ by this rule.
2. **A locked seed counts as not keeping up.** Decided before the evaluation, so the choice cannot follow the results.
3. **20 test seeds.** The runs already trained store 6 in their settings, so the evaluation needs a small option in the code to play 20. With 20 seeds, 5 have the layout at 8 agents (200000, 200005, 200012, 200017, 200018). The count for 32 agents is running.
4. **The mechanism goes in the Implementation chapter**, with the figure, as in 2.1.

Left for you. Is the figure's text right, and is the figure better beside conflict resolution or beside the threats to validity? Every number here can be made again with `thesis-progress/scripts/workstation_l` and `results/limitations/workstation_lock/README.md` lists the commands.